Econstellar Compute Engine: Verification Run Report, 2026-07-09

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Run executed 2026-07-09T02:29:39.569Z (UTC)

Abstract

We report a complete verification run of the Econstellar compute engine, a continuously deployed econometric service for financial-market analysis in which every public method ships with a pre-registered, failable evaluation. The suite executed 32 method rows — spanning stationarity, unit roots and cointegration; volatility, long memory and spectra; information flow and transfer entropy; connectedness and networks; contagion channels and systemic risk; reproduction and formal verification — against the live service, each checked against an expected band fixed before the run and naming its full parameterisation and source: 32 passed, 0 failed, 0 pending. Every quantity traces to a live computation or a documented published result; the methods execute the original papers' own software rather than reimplementations; and the document is generated mechanically from the run artefact, so that it may be regenerated and audited rather than taken on trust. Honest negatives, where present, are reported as the registered expectation: an empty network or a non-significant test is a result, not a blemish to manage.

Keywords: reproducible econometrics; transfer entropy; financial contagion; systemic risk; sandboxed computation; continuous verification.

JEL classification: C55, C58, C63, C88, G15.

1 The engine and the run

The engine is a sandboxed R/Node compute kernel on Google Cloud Run (service `shssm-compute`, region asia-south1), backed by an always-on workstation job server for long-running bootstrap and surrogate work. Each method executes the published package of its source paper rather than a reimplementation. The run below was produced by `evals/run-evals.mjs`, whose output `evals.json` is committed unedited beside the public pages and is the sole source of every number in this report.

Run timestamp	2026-07-09T02:29:39.569Z
Engine URL	`https://shssm-compute-b7ui3oxaqq-el.a.run.app`
Engine reachable at probe	yes
Engine revision	`shssm-compute-00055-bjz`
Registered methods	32
Suite rows	32
Summary	32 pass / 0 fail / 0 pending
Engine repo commit	`ad74104`
Pages repo commit	`bf585e2`

2 Data and methods

The engine analyses daily log-returns of major markets. Two panels recur: `g20` (18 markets, 5036 daily observations, 306 directed pairs) and `g20_24` (24 series, adding commodities, 552 directed pairs). Information-flow estimators require stationary inputs, so returns — integrated of order zero — are used throughout, never price levels. The 32 method rows group into the families below; each names the published basis it runs.

Stationarity, unit roots and cointegration. Augmented Dickey–Fuller and KPSS tests, the Im–Pesaran–Shin and Levin–Lin–Chu panel unit-root tests, and Johansen's trace test for cointegrating rank. The downstream information-flow estimators require inputs integrated of order zero, so log-returns are tested, never price levels.
Methods: `unit_root`, `live_unit_root`, `panel_unit_root`, `vecm`.

Volatility, long memory and spectra. GARCH(1,1) persistence, the Hurst exponent by detrended fluctuation analysis, and a maximal-overlap discrete wavelet transform for scale-resolved variance and cross-market coherence.
Methods: `garch`, `dfa_hurst`, `namh_hurst`, `wavelet`, `wavelet_coherence`.

Information flow and transfer entropy. Transfer entropy [3] estimated by the Kraskov–Stögbauer–Grassberger nearest-neighbour scheme [1] in the Frenzel–Pompe conditional-mutual-information form [2], with significance assessed against IAAFT surrogates [4] and edges selected under Benjamini–Hochberg false-discovery control [5]; a wavelet–quantile variant resolves spillover by scale and quantile.
Methods: `ksg_te`, `ksg_robustness`, `namh_te`, `namh_pipeline`, `wqte`, `soch_profile`, `quantile_var`.

Connectedness and networks. Vector-autoregressive impulse responses, Granger-causal edges, and the Diebold–Yılmaz forecast-error-variance connectedness index [6], static and rolling, with a dynamic conditional-correlation companion and an `igraph` network surface.
Methods: `var_irf`, `granger`, `connectedness`, `spillover_rolling`, `network`, `rolling_dcc`.

Contagion channels and systemic risk. Attribution of crisis-window spillovers to economic channels (trade, finance and others; the global-financial-crisis decomposition is anchored to arXiv:2604.26546), and a daily systemic-risk index recomputed in full from the stored panel.
Methods: `channel_attribution`, `sri_daily`.

Reproduction and formal verification. A delta-verifier that re-runs the published estimators of the `namh` package [8] and reports the maximum absolute deviation from the archived output, and a seeded robustness badge [7] over the published SOCH symmetry test.
Methods: `namh_reproduce`, `soch_robustness`.

3 Results

Table 1 groups the 32 rows by family, placing each verified value beside the pre-registered band it was checked against: 32 of 32 fall within band (0 fail, 0 pending). The bands are not decorative. Several encode honest negatives: where the engine ships an empty false-discovery-controlled network or a bit-exact reproduction target, the null or the exact value is the correct outcome, and a populated network or a moved number would be the failure, not the pass.

Table 1: Verification suite grouped by method family: the verified value, the pre-registered band with its parameterisation, and pass/fail status. Rendered verbatim from `evals.json`.

Method	Value	Pre-registered band (parameterisation)	Status
Stationarity, unit roots and cointegration			
`unit_root`	-49.17960762	India ADF stat in [-55,-45] (verified -49.18)	pass
`live_unit_root`	—	live S&P: levels non-stationary, returns stationary (the stationarity gate)	pass
`panel_unit_root`	-77.25938898	panel stationarity rejected: IPS and LLC ¡ -10, p $\approx$ 0 (this exact 5-market panel: IPS -77.26, LLC -51.79)	pass
`vecm`	3	Johansen rank 3 (verified trace 7265.97→1851.19)	pass
Volatility, long memory and spectra			
`garch`	0.99102203	India GARCH persistence in [0.95,1.0] (verified 0.991)	pass
`dfa_hurst`	0.54197024	India Hurst in [0.45,0.65] (verified 0.542)	pass
`namh_hurst`	0.48990576	g20_24 Gold H_mean in [0.4889,0.4909] (bit-exact verified 0.4899)	pass
`wavelet`	47.07	India d1 share of variance in [35,60]% (verified 47.07)	pass
`wavelet_coherence`	0.24851479666666665	mean coherence in [0.2,0.3] (verified 0.249, peak d6)	pass
Information flow and transfer entropy			
`ksg_te`	0.154234	USA→Japan full-sample KSG TE in [0.150,0.158] (verified 0.1542)	pass
`ksg_robustness`	0.7282	USA→Japan rank-1 in 8/8 (k,lag) configs; mean Spearman in [0.6,0.85] (verified 0.728)	pass
`namh_te`	-0.22322165	g20_24 window-1 te_mean in [-0.2242,-0.2222] (bit-exact verified -0.223222)	pass
`namh_pipeline`	-0.22322165	seeded end-to-end run (seed 42, L'Ecuyer-CMRG): FDR retains 0/552 under the paper's own gate (degenerate network, honest amber); raw te_mean in [-0.2242,-0.2222]	pass
`wqte`	0.03910699	USA→India tau.05 aggregate spillover in [0.02,0.06] (verified 0.039)	pass
`soch_profile`	0.03910699	USA→India aggregate in [0.03,0.05] (verified 0.0391, peak d4)	pass
`quantile_var`	0.7006	USA is the top tail driver with positive net score (verified +0.70)	pass
Connectedness and networks			
`var_irf`	0.70532304	stable VAR: max companion root ¡ 1, finite	pass
`granger`	6	6 significant directed edges (verified; USA out-degree 3)	pass
`connectedness`	30.25	total connectedness in [25,35]% (verified TCI 30.25)	pass
`spillover_rolling`	28.39	mean rolling TCI in [20,35]% (verified 28.4)	pass
`network`	0.5667	6-node graph, density in (0,1], communities $\geq$ 1	pass
`rolling_dcc`	0.2295	India–USA mean dynamic correlation in [0.15,0.31] (verified 0.23)	pass
Contagion channels and systemic risk			
`channel_attribution`	27.9	GFC: Trade 27.9% (Table 5, 0.000pp) and dominant == Trade	pass

Method	Value	Pre-registered band (parameterisation)	Status
`sri_daily`	0.00849	static-panel SRI = 0.00849 exactly ±1e-5 (first honest point, 306 pairs)	pass
Reproduction and formal verification			
`namh_reproduce`	1.94289029e-15	hurst GREEN: max—Δ— ≤ 1e-8 over ≥400 finite cells, 0 finite/NA mismatches; te GREEN ≤ 1e-8; fdr_network AMBER with 0/552 retained (D1: never green)	pass
`soch_robustness`	0.9911	SOCH-B badge, seeded tuple {seed 42 L'Ecuyer-CMRG, B=200, la8, advanced-8 = 28 pairs, grid tau{0.05,0.10}xJ{4,5} = 112 points}: baseline tau=0.05/J=4 holds 28/28 (paper ground truth reproduced) + pass_rate 0.9911 exact (111/112, matches Phase-31 0.991) -¿ badge robust	pass
Other			
`turnpike`	0.9675	turnpike eta-bar == 0.9675 == V.1 interior optimum; saddle with 1-D stable manifold (V.2 T3)	pass
`fragility_barrier`	1.25	feasibility barrier beta*r(B)=1 (barrier 1.25); flip onset analytic==numeric (2.745); positive-Lyapunov chaotic band (V.2 T4)	pass
`lq_regulator`	0.4237576593095	uncontrolled non-stationary (rho_open 1.207); optimal feedback stabilises (rho_closed 0.42376); cost ranking holds; certainty-equivalence residual 0 (V.2 T1)	pass
`bellman_value`	0.9675	Bellman contraction modulus == beta (0.95) in 406 iters; policy rest point == turnpike 0.9675 (V.2 T2)	pass
`regime_conditioned_te`	0.100481	crisis/D1 anchor tuple: 535 regime days, 12 pairs, top edge S&P 500 -¿ Nikkei 225 with te in [0.0995,0.1015] (observed TE deterministic; significance not banded)	pass
`news_attention_te`	0.065781	TE(inflation→election) = 0.065781 (6dp exact; deterministic CPU reproduces the published GPU TE_matrix)	pass

4 Failures and pending rows

All 32 rows fall within their pre-registered bands. No band was widened and no extraction was altered after the run; the tables here are the unedited rendering of a single execution of the committed runner.

5 Reproducibility and integrity

This report is generated mechanically from the run artefact and inherits its discipline. (i) Every row carries a pre-registered band naming its full parameterisation and a source, fixed before the run. (ii) `evals.json` is the output of one genuine run of the committed runner and is never hand-edited; to change a number one re-runs the suite, not the report. (iii) A band failure halts work — bands are never widened to pass, and only extraction plumbing (the harness misreading a result shape) may be calibrated, with the band untouched. (iv) Honest negatives are permanent: a non-significant test is reported as non-significant, an empty false-discovery network as empty [5], an unproved formal statement as unproved. (v) Asynchronous rows are real workstation jobs with persistent identifiers, or are reported pending — never fabricated.

Reproduction is graded rather than asserted. Deterministic estimators — the nearest-neighbour transfer entropy [1] and the Hurst panel of the published `namh` package [8] — are reproduced to a stated maximum absolute deviation and earn a green only on an exact match; stochastic surrogate stages, whose generators were unseeded in the original runs, are reported amber rather than dressed as exact. Where the engine seeds its own surrogate and bootstrap stages it uses a single declared generator [7], so a result is reproducible for a fixed seed and core count and the asymmetry with the unseeded published run is stated, not concealed. The engine and this verification apparatus are described in full in the accompanying system paper [9].

A Per-row provenance

Method	Source band / provenance	Tower job
`unit_root`	A2 + golden	sync
`var_irf`	A2 invariant (verified tuple: lag 7, root 0.705)	sync
`dfa_hurst`	A2 + golden	sync
`garch`	A2 + golden	sync
`wavelet`	A2 + golden	sync
`wqte`	A2	sync
`soch_profile`	A2	sync
`channel_attribution`	A2 + arXiv:2604.26546	sync
`namh_hurst`	A2 #45	sync
`namh_te`	A2 #45	sync
`vecm`	A2	sync
`granger`	A2	sync
`panel_unit_root`	A2 invariant — series set resolved 2026-06-11: the documented tuple is the {India,USA,UK,China,Japan} panel (live re-run -77.2594/-51.7933, exact)	sync
`connectedness`	A2	sync
`network`	A2 invariant	sync
`rolling_dcc`	A2	sync
`wavelet_coherence`	A2	sync
`spillover_rolling`	A2	sync
`quantile_var`	A2	sync
`live_unit_root`	A2 + invariant 4 (TE on I(0) returns, never levels)	sync
`sri_daily`	A2 Phase 30	sync
`ksg_te`	A2 + #39 anchor	`job_20260709_b0c99a09`
`ksg_robustness`	A2 G.3	`job_20260709_c043deba`
`namh_reproduce`	M1 Step 5 — measured 2026-06-12: same-machine 5.0e-16 (440 cells) / 1.7e-16; live-vs-cache ≈4.9e-9 cross-BLAS (#45/#46)	sync
`namh_pipeline`	M1 Step 4 — determinism verified 2026-06-12 (same seed → identical surrogate p-values; different seed → different); FDR-empty = the published result (#46)	`job_20260709_19b3f62b`
`soch_robustness`	Phase 31 closure board + seeded anchor run job_20260612_fe0c4f06 (2026-06-12; an unseeded run flipped the baseline pair once, 27/28 — driver got the namh_pipeline seed pin, band never widened)	`job_20260709_842ea2f3`
`turnpike`	v5-control.test.mjs GOLD.tp + V.2 results.json	sync
`fragility_barrier`	v5-control.test.mjs GOLD.fb + V.2 results.json	sync
`lq_regulator`	v5-control.test.mjs GOLD.lq + V.2 results.json	sync
`bellman_value`	v5-control.test.mjs GOLD.vi + V.2 results.json	sync
`regime_conditioned_te`	anchor run job_20260704_63f7a48f (tower, mstcontagion pkg) 2026-07-04	`job_20260709_d20a1d87`

Method	Source band / provenance	Tower job
`news_attention_te`	A2 + news_attention_te.R method desc (FRONTIERS III)	sync

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